

The Complete Block-System Solution and a Compatibility Correction for the Principal Pivot Transform

A full correction of Problem 7 and Proposition 8 in arXiv:2302.04293

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Abstract

We solve the block linear system posed as Problem 7 of Beard–Welters for an arbitrary real or complex block matrix. The claimed solution in their Proposition 8 omits a necessary range-compatibility condition and is false even for the zero matrix. We give the complete affine solution, repair the proposition sharply, and identify the projected system actually encoded by the generalized principal pivot transform when compatibility fails.

1 Source problem and the missing condition

Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad A_{ij} \in M_{n_i \times n_j}(\mathbb{F}), \quad \mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}.$$

Problem 7 of Beard–Welters asks, for prescribed (x_1, y_2) , to find every (y_1, x_2) such that

$$A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}. \quad (1)$$

Their Proposition 8 claims a solution under $\text{ran } A_{21} \subseteq \text{ran } A_{22}$ and $\ker A_{22} \subseteq \ker A_{12}$, for every prescribed (x_1, y_2) .

Problem 7 *Let $A = [A_{ij}]_{i,j=1,2} \in M_n(\mathbb{F})$. Given $(x_1, y_2) \in \mathbb{F}^{n_1} \times \mathbb{F}^{n_2}$, find all $(y_1, x_2) \in \mathbb{F}^{n_1} \times \mathbb{F}^{n_2}$ such that*

$$A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}. \quad (21)$$

Proposition 8 (Solution of Problem 7) *If $A = [A_{ij}]_{i,j=1,2} \in M_n(\mathbb{F})$, $\text{ran } A_{21} \subseteq \text{ran } A_{22}$, and $\ker A_{22} \subseteq \ker A_{12}$ then for each $(x_1, y_2) \in \mathbb{F}^{n_1} \times \mathbb{F}^{n_2}$ and for $(y_1, x_2) \in \mathbb{F}^{n_1} \times \mathbb{F}^{n_2}$,*

$$A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \iff \begin{cases} x_2 \in \{-A_{22}^+ A_{21} x_1 + A_{22}^+ y_2\} + \ker A_{22}, \\ y_1 = A/A_{22} x_1 + A_{12} A_{22}^+ y_2 \end{cases} \quad (22)$$

with a particular solution $(y_1, x_2^0) \in \mathbb{F}^{n_1} \times \mathbb{F}^{n_2}$ given by

$$J \text{ppt}(A) \begin{bmatrix} x_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ -x_2^0 \end{bmatrix}, \quad x_2^0 = -A_{22}^+ A_{21} x_1 + A_{22}^+ y_2. \quad (23)$$

The omitted issue is that the second block equation need not be consistent. Applying A_{22}^+ to it does not retain the component perpendicular to $\text{ran } A_{22}$.

2 Idea of Proof

Put $b = y_2 - A_{21}x_1$. The second block row of (1) is exactly $A_{22}x_2 = b$. If $P = A_{22}A_{22}^+$, then P is the orthogonal projection onto $\text{ran } A_{22}$, so this equation is solvable precisely when $(I - P)b = 0$. In that event all solutions are $x_2 = A_{22}^+b + z$ with $z \in \ker A_{22}$. Substitution in the first block row gives all and only the corresponding y_1 .

3 Complete solution for arbitrary blocks

Theorem 1 (Complete solution of Problem 7). *For arbitrary A and prescribed (x_1, y_2) , set*

$$b = y_2 - A_{21}x_1, \quad P = A_{22}A_{22}^+.$$

There exists a pair (y_1, x_2) satisfying (1) if and only if any, hence all, of the equivalent conditions hold:

$$b \in \text{ran } A_{22}, \quad Pb = b, \quad (I - P)(y_2 - A_{21}x_1) = 0.$$

When they hold, the complete solution set is

$$\left\{ \begin{pmatrix} y_1 = A_{11}x_1 + A_{12}A_{22}^+b + A_{12}z, \\ x_2 = A_{22}^+b + z \end{pmatrix} : z \in \ker A_{22} \right\}. \quad (2)$$

Equivalently, with $A/A_{22} = A_{11} - A_{12}A_{22}^+A_{21}$,

$$y_1 = (A/A_{22})x_1 + A_{12}A_{22}^+y_2 + A_{12}z.$$

Proof. The second block row is $A_{22}x_2 = b$. The Moore–Penrose identities imply that $P = A_{22}A_{22}^+$ is the orthogonal projection onto $\text{ran } A_{22}$. Thus solvability is equivalent to $Pb = b$. If this holds, $A_{22}(A_{22}^+b) = b$, and adding any $z \in \ker A_{22}$ preserves the equation. Conversely, if $A_{22}x_2 = b$, then $x_2 - A_{22}^+b$ lies in $\ker A_{22}$. This proves the asserted complete formula for x_2 . The first block row uniquely sets $y_1 = A_{11}x_1 + A_{12}x_2$; substituting the formula for x_2 gives (2) and its Schur-complement form. $\square$

4 Counterexample and sharp repair of Proposition 8

Proposition 2 (Minimal counterexample). *Proposition 8 of the source is false as stated.*

Proof. Take $n_1 = n_2 = 1$, $A = 0$, $x_1 = 0$, and $y_2 = 1$. Then $\text{ran } A_{21} = \text{ran } A_{22} = \{0\}$ and $\ker A_{22} = \ker A_{12} = \mathbb{F}$, so both hypotheses of the proposition hold. Its right-hand side (22) says $x_2 \in \mathbb{F}$ and $y_1 = 0$, hence predicts many solutions. But

$$A \begin{bmatrix} 0 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

for every x_2 . Equivalently, the missing condition is $1 \in \text{ran } A_{22} = \{0\}$, which fails. $\square$

Corollary 3 (Corrected Proposition 8). *Assume the source hypotheses*

$$\text{ran } A_{21} \subseteq \text{ran } A_{22}, \quad \ker A_{22} \subseteq \ker A_{12}.$$

For prescribed (x_1, y_2) , a solution exists if and only if $y_2 \in \text{ran } A_{22}$. When it exists, every solution is

$$\begin{aligned} x_2 &\in \{-A_{22}^+A_{21}x_1 + A_{22}^+y_2\} + \ker A_{22}, \\ y_1 &= (A/A_{22})x_1 + A_{12}A_{22}^+y_2. \end{aligned}$$

Thus the source formula becomes correct after adding exactly the condition $y_2 \in \text{ran } A_{22}$.

Proof. Since $\text{ran } A_{21} \subseteq \text{ran } A_{22}$, the vector $A_{21}x_1$ already lies in $\text{ran } A_{22}$; hence $y_2 - A_{21}x_1 \in \text{ran } A_{22}$ iff $y_2 \in \text{ran } A_{22}$. Theorem 1 gives the stated affine formula. Finally $\ker A_{22} \subseteq \ker A_{12}$ makes $A_{12}z = 0$, so y_1 is independent of the kernel parameter. $\square$

5 What the principal pivot transform always encodes

For arbitrary A and data define

$$x_2^0 = A_{22}^+(y_2 - A_{21}x_1), \quad y_1^0 = A_{11}x_1 + A_{12}x_2^0.$$

Direct multiplication, with no range hypotheses, gives the valid identity

$$J_{\text{ppt}}(A) \begin{bmatrix} x_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} y_1^0 \\ -x_2^0 \end{bmatrix}.$$

However, the associated block system is

$$A \begin{bmatrix} x_1 \\ x_2^0 \end{bmatrix} = \begin{bmatrix} y_1^0 \\ A_{21}x_1 + P(y_2 - A_{21}x_1) \end{bmatrix}, \quad (3)$$

not necessarily (1). Under $\text{ran } A_{21} \subseteq \text{ran } A_{22}$ the lower entry in (3) is simply Py_2 . Hence the source's transform always solves the projected-data system and solves the originally prescribed system exactly when the compatibility condition holds.

This correction concerns the claimed linear-system interpretation. The source's later variational formula already contains the projection $A_{22}A_{22}^+$ explicitly, so no broader invalidity is asserted here.

6 Verification and novelty scope

The accompanying script checks compatible and incompatible random singular block systems, the complete affine formula, the minimal counterexample, and the projected-data identity. A bounded search found the current arXiv v2 source but no erratum or later correction adding the missing compatibility condition. The novelty claim is limited to this correction and complete solution of the displayed Problem 7.

References

- [1] K. Beard and A. Welters, *Matrix monotonicity and concavity of the principal pivot transform*, arXiv:2302.04293v2 (2023), Problem 7 and Proposition 8.